

MT AIRY, NC, USA

WMO: 723177

Lat: 36.460N

Lon: 80.553W

Elev: 1247

StdP: 14.05

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 63807

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.7	22.8	-8.1	3.7	32.9	-1.9	5.2	32.5	17.7	39.2	15.8	40.7	3.3	320	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	91.4	74.0	90.0	73.6	87.9	73.3	78.1	86.9	76.8	85.4	75.5	83.9	4.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	138.8	83.3	74.0	132.6	82.0	72.9	127.9	80.8	42.6	87.7	41.1	85.4	40.0	83.5	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	12.2	10.5	13.0	95.1	4.7	2.9	9.6	97.2	6.8	98.9	4.2	100.5	0.7	102.6
			10.3	79.4	4.2	1.5	7.2	80.5	4.8	81.4	2.4	82.3	-0.6	83.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.0	39.3	42.1	49.9	58.9	66.4	74.2	77.5	76.5	70.5	59.6	49.6	42.9
	DBStd	15.28	9.49	8.71	8.86	7.75	6.87	4.76	3.93	4.41	5.58	7.69	8.32	7.87
	HDD50	1097	353	247	115	16	0	0	0	0	12	107	247	
	HDD65	3561	798	642	472	209	71	4	0	0	18	199	463	685
	CDD50	4399	20	26	111	283	509	727	851	823	615	311	96	27
	CDD65	1387	0	1	3	27	114	280	386	358	183	33	2	0
	CDH74	11435	2	2	40	324	907	2323	3371	2970	1227	243	26	0
	CDH80	3902	0	0	1	59	229	804	1321	1105	339	43	1	0
(14) Wind	WSAvg	3.1	3.6	3.8	4.4	4.1	3.1	2.5	2.0	2.1	2.8	2.9	3.1	3.0
(15) Precipitation	PrecAvg	48.2	3.7	3.4	4.4	3.8	4.4	4.3	4.9	4.4	4.3	3.5	3.4	3.5
	PrecMax	70.0	7.7	7.3	9.8	8.9	8.3	10.0	14.3	12.5	11.3	11.2	9.3	8.3
	PrecMin	32.3	0.8	0.5	0.8	0.6	1.2	0.7	1.5	0.5	0.2	0.0	0.6	0.3
	PrecStd	7.8	1.7	1.7	2.0	1.9	1.8	2.4	2.3	2.8	2.6	2.4	1.7	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.9	71.9	79.0	85.7	89.8	93.0	94.9	95.4	90.2	84.5	78.6	70.4
		MCWB	57.9	60.5	59.0	61.7	70.6	73.2	75.7	73.4	72.0	72.3	61.8	58.7
	2%	DB	63.4	66.2	73.4	81.3	84.5	90.2	91.4	91.0	86.8	79.9	72.4	64.1
		MCWB	54.4	56.3	57.4	61.9	69.3	73.0	75.3	73.7	72.1	65.7	58.8	56.4
	5%	DB	59.5	62.3	70.1	78.1	82.2	87.8	89.9	89.5	83.9	76.4	68.3	60.6
		MCWB	51.4	51.0	55.7	61.6	68.4	72.5	74.7	73.6	71.2	64.7	57.8	53.4
	10%	DB	54.6	57.1	65.3	73.5	79.5	84.4	87.6	86.2	81.3	72.8	64.2	55.8
		MCWB	47.3	47.0	53.4	59.0	66.8	71.3	74.2	73.9	69.1	63.5	55.2	48.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.1	63.6	64.1	69.7	75.1	77.5	79.9	80.2	76.5	74.5	66.5	64.9
		MCDB	65.4	69.5	73.2	76.0	84.1	86.8	88.6	89.6	84.6	81.1	72.0	67.7
	2%	WB	57.2	59.1	61.2	66.4	72.6	76.1	78.2	78.2	75.2	71.2	63.5	59.6
		MCDB	62.2	63.8	68.2	74.9	81.1	85.2	87.4	86.6	83.3	76.2	67.7	62.6
	5%	WB	52.9	53.3	59.1	64.3	70.8	74.9	77.0	76.7	73.6	68.0	60.2	54.6
		MCDB	57.2	59.4	66.7	73.3	79.2	83.8	86.2	84.5	80.6	73.8	66.2	58.1
	10%	WB	48.3	48.6	56.3	62.3	69.0	73.5	75.7	75.2	71.9	64.9	56.8	50.5
		MCDB	53.2	54.8	63.5	71.0	76.9	81.9	84.3	82.7	77.7	70.4	62.8	54.6
(35) Mean Daily Temperature Range	5% DB	MDBR	17.8	18.5	20.1	21.7	19.5	19.0	17.8	17.3	17.4	19.4	20.3	16.7
		MCDBR	24.2	25.9	26.1	27.4	23.1	22.2	20.9	20.9	21.6	23.8	24.9	22.1
		MCWBR	16.1	16.8	13.5	12.6	11.1	9.3	8.6	8.3	10.0	12.3	13.8	14.9
	5% WB	MCDBR	19.7	22.4	21.8	21.2	19.3	19.0	18.3	17.3	17.4	18.0	18.2	17.5
		MCWBR	16.2	18.0	14.1	11.9	10.5	9.4	8.9	8.5	9.3	11.5	12.9	15.0
(40) Clear-Sky Solar Irradiance	taub		0.304	0.317	0.344	0.390	0.430	0.466	0.504	0.504	0.401	0.359	0.318	0.304
	taud		2.493	2.450	2.420	2.304	2.210	2.147	2.068	2.043	2.330	2.434	2.505	2.527
	Ebn at Noon		284	292	293	283	272	261	250	247	270	273	276	277
	Edh at Noon		26	30	34	41	45	49	52	52	37	30	25	23
(44) All-Sky Solar Radiation	RadAvg		783	989	1327	1654	1819	1960	1819	1673	1407	1151	872	666
	RadStd		55	119	82	135	150	144	117	87	132	141	73	62

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (51 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	+157

Nomenclature: See separate page